

GATE 2027 CS Study Guide

Prepared 28 September 2026

GATE 2027 runs 6–21 February 2027, which gives you about 18 weeks from today. Register first: the no-late-fee deadline was extended to 5 October 2026, and the late-fee window closes 12 October 2026.

Exam at a glance

Item	Detail
Organiser	IIT Madras (official site)
Exam dates	6, 7, 13, 14, 20, 21 Feb 2027; CS date and session not yet announced. Plan for the earliest, 6 Feb 2027
Sessions	Forenoon 9:30–12:30, afternoon 2:30–5:30
Results	19 March 2027
Paper	3 hours, computer-based, 65 questions, 100 marks
Split	General Aptitude 15 marks (10 Qs) + CS incl. Engineering Maths 85 marks (55 Qs)
Question types	MCQ, MSQ (multiple correct), NAT (type a number)
Negative marking	MCQ only: $-1/3$ on 1-mark, $-2/3$ on 2-mark. MSQ and NAT have none
Calculator	On-screen virtual calculator only; practise with it

Sources: [IIT Madras press release](#), [registration extension](#), [session timings](#), [marking scheme](#).

Syllabus analysis: 2026 vs 2027

Study from the GATE 2027 syllabus, not 2026. Eight of the ten sections are word-for-word the same; Digital Logic and COA gained design depth, and Computer Networks was cut sharply. General Aptitude (verbal, quantitative, analytical, spatial) is unchanged.

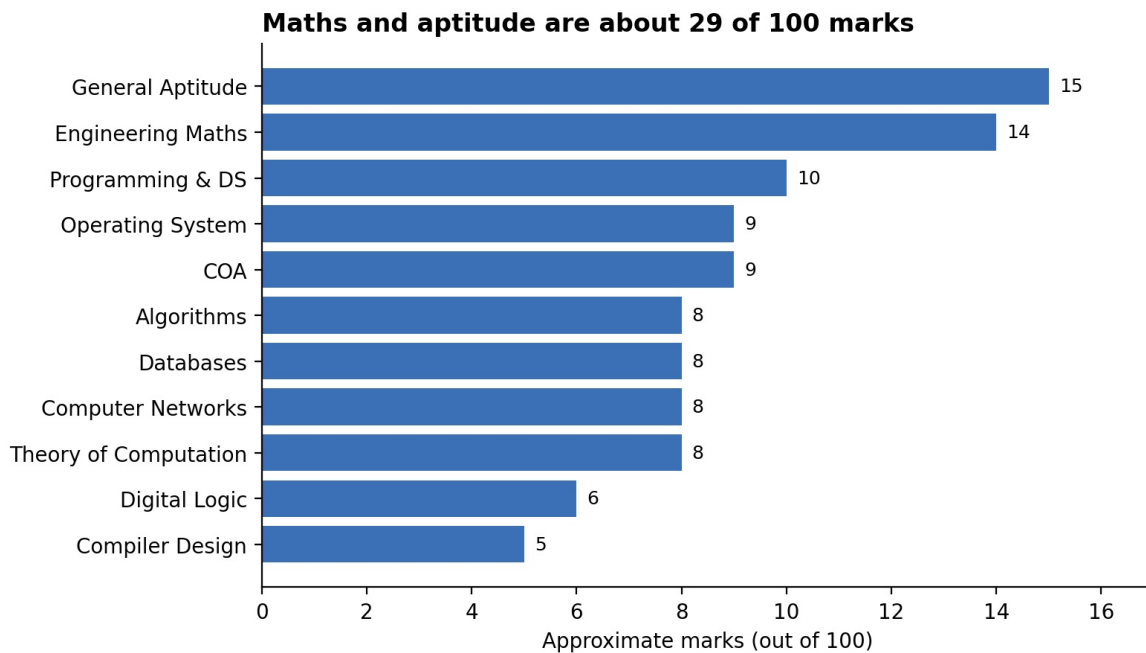
#	Section	GATE 2027 topics	Change vs 2026
1	Engineering Maths	Discrete maths (logic, sets, relations, functions, posets, lattices, monoids, groups, graphs, combinatorics, recurrences, generating functions); linear algebra (matrices, determinants, linear systems, eigenvalues, LU); calculus (limits, continuity, maxima-minima, MVT, integration); probability and statistics (distributions, mean/median/mode/SD, Bayes)	Same
2	Digital Logic	Boolean algebra and minimisation (algebraic, K-map, tabular/Quine-McCluskey); design of combinational and sequential circuits; fixed and floating point	Now names K-map and tabular method explicitly; asks for circuit design
3	COA	Instruction set, addressing modes; ALU design; control unit design (hardwired, microprogrammed); memory interfacing, hierarchy performance, cache mapping; interrupt and DMA; pipelining and hazards	Added control-unit design and memory interfacing; secondary storage dropped
4	Programming & DS	C, recursion, arrays, stacks, queues, linked lists, trees, BSTs, binary heaps, graphs	Same
5	Algorithms	Searching, sorting, hashing; asymptotic complexity; greedy, DP, divide and conquer; graph traversals, MST, shortest paths	Same
6	Theory of Computation	Regex and FA; CFG and PDA; regular and CFLs, pumping lemma; Turing machines, undecidability	Same
7	Compiler Design	Lexing, parsing, SDT; runtime environments; intermediate code; local optimisation; constant propagation, liveness, CSE	Same
8	Operating System	System calls, processes, threads, IPC, synchronisation; deadlock; CPU and I/O scheduling; memory management, virtual memory; file systems	Same
9	Databases	ER model; relational algebra, tuple calculus, SQL; constraints, normal forms; file organisation, B and B+ tree indexing; transactions, concurrency control	Same
10	Computer Networks	Layering; circuit, packet and virtual-circuit switching + performance metrics; error detection, MAC, Ethernet; distance-vector and link-state routing; IPv4 fragmentation, CIDR, NAT; TCP flow and	Trimmed: framing, bridging, flooding, ARP/DHCP/ICMP, UDP, SMTP, FTP and email removed;

What this means for you: skip the removed CN topics, spend extra time on CN numericals (delays, throughput, window sizes), and practise control-unit and cache-mapping questions in COA.

Sources: [GATE 2027 CS syllabus \(IIT Madras\)](#), [GATE 2026 CS syllabus \(IIT Guwahati\)](#).

Weightage and priority

No subject is skippable, but GA, Maths and Programming + Algorithms together carry about half the paper. Figures below are approximate averages from past-paper analyses, not official.



Approximate marks per subject

Priority tiers for a 4-month plan

- **Tier 1 — highest return per hour:** General Aptitude, Engineering Maths (especially discrete maths and probability), Programming & DS + Algorithms (these two alone gave 16–18 marks in 2023–2025).
- **Tier 2 — steady, predictable marks:** Operating System, COA, Databases, Computer Networks, Theory of Computation.
- **Tier 3 — small but quick to finish:** Digital Logic, Compiler Design. Each takes about a week and questions repeat patterns closely.

Sources: [GO Classes weightage analysis 2023–2025](#), [PW subject-wise weightage](#).

19-week plan

The plan has three phases: learn every subtopic by 10 Jan (weeks 1–15), revise with subject tests (week 16), then full mocks until the exam (weeks 17–19). It assumes about 6 focused hours a day, 6 days a week, with Sunday for catch-up and a weekly test.

Every day, on top of the week's subject: 30 minutes of General Aptitude, and 1.5 hours of previous-year questions (PYQs) on whatever you studied that day.

Week	Starts	Main subject	Finish-line check
1	28 Sep	Register for GATE. Discrete Maths I: logic, sets, relations, posets, lattices, functions	Logic + set theory PYQs
2	5 Oct	Discrete Maths II: groups, graphs, counting, recurrences, generating functions	All discrete maths PYQs
3	12 Oct	Programming in C, recursion, stacks, queues	Trace 30 C output questions by hand
4	19 Oct	Linked lists, trees, BST/AVL, heaps, graphs, hashing	DS PYQs done

5	26 Oct	Algorithms I: asymptotics, recurrences, searching, sorting, divide and conquer, greedy	Master theorem without notes
6	2 Nov	Algorithms II: dynamic programming, graph traversals, MST, shortest paths	Algorithms PYQs done
7	9 Nov	OS I: processes, threads, scheduling, synchronisation, deadlock	Gantt charts + semaphore PYQs
8	16 Nov	OS II: memory, paging, virtual memory, file systems. Plus linear algebra	OS + linear algebra PYQs
9	23 Nov	Calculus + Digital Logic I: number representation, floating point, minimisation, combinational design	Digital logic PYQs
10	30 Nov	Digital Logic II: sequential design. COA I: ISA, addressing, control unit, memory, cache	Cache numericals timed
11	7 Dec	COA II: I/O, DMA, pipelining. Probability. Databases I: ER, relational algebra	Pipeline + probability PYQs
12	14 Dec	Databases II: SQL, FDs, normal forms, indexing, B+ trees, transactions	DBMS PYQs done
13	21 Dec	Computer Networks (2027 syllabus only)	Subnetting, window size, delay numericals
14	28 Dec	Theory of Computation	Closure/decidability table from memory
15	4 Jan	Compiler Design; full diagnostic mock on 10 Jan	Every subtopic covered once
16	11 Jan	Revision: two subjects a day + subject tests	Scoring 60%+ on subject tests
17	18 Jan	Full mocks + analysis, weak-topic fixes	Each mock reviewed within 24 hours
18	25 Jan	GATE 2024–2026 papers as timed mocks	Mock scores trending up
19	1 Feb	Short-note revision, rest the day before	Calm and slept on exam eve

If the CS paper lands on 13–21 Feb, stretch weeks 17–19 rather than learning anything new. The day-by-day version is in Part 3.

Study material by subject

Use one lecture series to learn, one standard book to look things up, and PYQs to test. With 18 weeks you won't read the books cover to cover; open them only for chapters where the lectures felt thin. Playlist links are in Part 2.

Subject	Learn from (free video)	Reference book	Topics that repeat in GATE
General Aptitude	Solve past GATE GA sections	R.S. Aggarwal, <i>Quantitative Aptitude</i>	Ratios, percentages, data interpretation, paper folding, analogies
Discrete Maths	Gate Smashers	Kenneth Rosen, <i>Discrete Mathematics and Its Applications</i>	Predicate logic, POSET/lattice, group properties, graph colouring, counting
Linear Algebra & Calculus	GO Classes; GATE Wallah	Strang, <i>Introduction to Linear Algebra</i>	Eigenvalues, rank, determinants, maxima-minima
Probability & Stats	GATE Wallah	Sheldon Ross, <i>A First Course in Probability</i>	Bayes, expectation, binomial and Poisson
Digital Logic	Neso Academy	Morris Mano, <i>Digital Design</i>	K-maps, MUX functions, counters, IEEE 754 floats
COA	Gate Smashers	Hamacher, <i>Computer Organization</i>	Cache mapping, pipeline speedup, addressing modes, DMA
C & Data Structures	Neso Academy	Kernighan & Ritchie, <i>The C Programming Language</i>	Pointer and recursion output, tree traversals, heap operations
Algorithms	Abdul Bari	Cormen et al., <i>CLRS</i>	Recurrences, DP, Dijkstra/MST, sorting complexity
Theory of Computation	Gate Smashers; Neso Academy	Peter Linz, <i>Formal Languages and Automata</i>	Minimal DFA, closure properties, decidability
Compiler Design	Gate Smashers	Aho et al., <i>Compilers</i> (Dragon book)	FIRST/FOLLOW, LL(1)/LR tables, SDT, liveness
Operating System	Gate Smashers	Silberschatz (Galvin), <i>Operating System Concepts</i>	Scheduling, semaphores, Banker's, paging and TLB numericals
Databases	Gate Smashers	Korth (Silberschatz), <i>Database System Concepts</i>	Normal forms, SQL output, B+ tree order, serialisability

Computer Networks	Gate Smashers	Kurose & Ross, <i>Computer Networking</i>	Subnetting/CIDR, sliding window, TCP congestion window, delays
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Practice sources, in order of priority

1. [GATE Overflow](#) — every CS PYQ since 1987 with discussed solutions, sorted by topic. Your main practice bank.
2. Official past papers and answer keys on the GATE website — use the 2024–2026 papers as timed mocks late in the plan.
3. One paid test series for full mocks (for example GO Classes, Made Easy or Applied Roots). Pick one; the brand matters less than reviewing every mock.
4. The GATE mock-test interface on the official site, to get used to the virtual calculator.

Practice, mocks and revision

The score comes from how you review mistakes, not from how many questions you attempt.

- **Short notes as you go:** one page per topic, only formulas, traps and question patterns. These become your whole revision material in weeks 15–19.
- **Error log:** for every wrong or guessed question, write the topic, why you got it wrong (concept, calculation, misread), and the fix. Reread it every Sunday.
- **PYQs by topic first, by year later:** solve topic-wise from GATE Overflow while learning; save the last three years' full papers for timed mocks.
- **Mock review:** spend as long reviewing a mock as taking it. Re-solve every wrong question without looking at the solution.
- **Spaced revision:** revisit each subject roughly 1 day, 1 week and 1 month after finishing it, using your short notes.
- **MSQ practice:** MSQs have no partial credit, so train yourself to check every option independently.

Exam-day tactics

- **First pass (about 90 minutes):** do General Aptitude and every question you can solve confidently. Mark the rest.
- **Second pass:** long numericals and marked questions. NAT and MSQ carry no negative marks, so always answer them.
- **MCQ guesses:** guess only if you can eliminate at least two options; a blind guess costs 1/3 or 2/3 of a mark.
- **Calculator:** use the on-screen calculator only for arithmetic you can't do quickly by hand; it is slow.
- **Last 10 minutes:** check NAT answers for units and rounding, and make sure no NAT or MSQ is left blank.
- **Admit card:** download it in January and carry it with a valid photo ID; entry isn't allowed without it.

Part 2 — Video Playlists

Every free YouTube playlist you need, in the order of the plan. Open the link for the current week and work through it top to bottom. Gate Smashers teaches in Hindi mixed with English; Neso Academy and Abdul Bari teach in English.

Master order

Week	Subject	Main playlist	Backup
1-2	Discrete Maths	Gate Smashers – Discrete Mathematics	—
3-4	C + Data Structures	Neso – C Programming & Data Structures	Neso – Data Structures
5-6	Algorithms	Abdul Bari – Algorithms	Gate Smashers – DAA
7-8	Operating System	Gate Smashers – Operating System	—
8	Linear Algebra	GO Classes – Linear Algebra for GATE CSE	—
8, 10	Calculus, Probability	GATE Wallah – Engineering Maths crash course (watch only CS-syllabus topics)	—
9	Digital Logic	Neso – Digital Electronics	—
9-10	COA	Gate Smashers – Computer Organization & Architecture	—
11	Databases	Gate Smashers – DBMS	—
12	Computer Networks	Gate Smashers – Computer Networks (skip topics removed for 2027)	—
13	Theory of Computation	Gate Smashers – TOC	Neso – TOC & Automata
14	Compiler Design	Gate Smashers – Compiler Design	—

DSA topic by topic (optional deep dives)

Use these only when a topic from the main playlist feels weak. They are interview-style, so stop once the concept is clear.

Topic	Playlist
Recursion	Striver – Recursion & Backtracking
Stack	Aditya Verma – Stack
Heap	Aditya Verma – Heap
Trees	Striver – Trees
Graphs	Striver – Graphs
Dynamic Programming	Aditya Verma – DP

PYQ solutions

Solve each question yourself first, then check the solution.

Resource	What it covers
GATE Overflow	Every CS PYQ with written solutions, sorted by topic. Your main PYQ bank
GO Classes (YouTube channel)	PYQ and standard-question discussions by subject
GO Classes – Digital Logic questions	GATE-level Digital Logic problems solved
Neso – GATE Solved Problems, Digital Electronics	Past GATE Digital Logic questions
GATE CSE PYQ – Last 30 Years	Mixed-subject PYQ walkthroughs
GATE CS & IT – Previous Year Question Series	Mixed-subject PYQ walkthroughs

General Aptitude has no playlist here. Practise it straight from the GA sections of past GATE papers, 30 minutes a day.

Part 3 — Daily Planner: 28 Sep 2026 - 6 Feb 2027

One entry per day. Every subtopic GATE 2027 CS tests is covered by 10 January, followed by one revision week and three weeks of mocks. Tick each day when it's done; if you fall behind, use the next Sunday to catch up rather than shifting the whole plan.

Every topic is from the official [GATE 2027 CS syllabus \(IIT Madras\)](#). Topics removed for 2027 (UDP, ARP/DHCP/ICMP, SMTP/FTP, framing, bridging, secondary storage) are left out. Each day names the playlist to watch; open it and play the videos whose titles match that day's subtopics.

Every study day (about 6 hours): 3.5 h lecture + short notes on the day's topic → 1.5 h PYQs on those exact subtopics from GATE Overflow → 30 min General Aptitude (this week's GA focus is under each week heading) → 10 min updating your error log.

Every Sunday: a timed test on the week's PYQs, then review the error log and finish anything left over. Include one "what does this code compute?" question, and check every option of each MSQ separately.

Phase 1 — Learn the syllabus (28 Sep - 10 Jan)

Week 1 · 28 Sep - 4 Oct · Discrete Maths I: logic, sets, relations

GA focus (30 min daily): Percentages, ratio and proportion

✓	Day	Topic and what to cover	Watch / practise
☐	Mon 28 Sep	Register for GATE (deadline 5 Oct). Propositional logic: connectives, truth tables, tautology, contradiction, contingency; validity and satisfiability; logical equivalence laws; implication, converse, contrapositive; CNF and DNF	Gate Smashers Discrete Maths
☐	Tue 29 Sep	Rules of inference and first-order logic: modus ponens/tollens, resolution, checking valid arguments; predicates, universal and existential quantifiers, nested quantifiers, negating quantified statements, translating English to logic, free vs bound variables	Gate Smashers Discrete Maths
☐	Wed 30 Sep	Sets and relations: set operations, power set, Cartesian product, inclusion-exclusion on sets; relations as sets, matrix and digraph form; reflexive, irreflexive, symmetric, antisymmetric, asymmetric, transitive; composition, inverse	Gate Smashers Discrete Maths
☐	Thu 1 Oct	Equivalence relations and counting relations: closures (reflexive, symmetric, transitive, Warshall's algorithm); equivalence classes and partitions; counting reflexive, symmetric, antisymmetric and equivalence relations (Bell numbers)	Gate Smashers Discrete Maths
☐	Fri 2 Oct	Partial orders and lattices: posets, Hasse diagrams, maximal/minimal/greatest/least elements, upper and lower bounds, LUB/GLB; total order, well-order; lattices, sublattices, bounded, distributive and complemented lattices, Boolean algebra as a lattice	Gate Smashers Discrete Maths
☐	Sat 3 Oct	Functions: injective, surjective, bijective; counting functions, one-one and onto functions (Stirling numbers of the second kind); composition, inverse, floor and ceiling; pigeonhole with functions	Gate Smashers Discrete Maths
☐	Sun 4 Oct	Weekly test: timed PYQs: logic, sets, relations, posets, lattices, functions; update error log	GATE Overflow PYQs

Week 2 · 5 - 11 Oct · Discrete Maths II: groups, graphs, combinatorics

GA focus (30 min daily): Averages, mixtures and alligation

✓	Day	Topic and what to cover	Watch / practise
☐	Mon 5 Oct	Algebraic structures: closure, associativity, identity, inverse; semigroups, monoids, groups, abelian groups; order of a group and of an element; subgroups, cyclic groups and generators, Lagrange's theorem, cosets; group tables	Gate Smashers Discrete Maths
☐	Tue 6 Oct	Graph basics: simple graphs, multigraphs, degree, handshaking lemma, degree sequences; complete, bipartite, regular, complement graphs; subgraphs; isomorphism; counting labelled graphs	Gate Smashers Discrete Maths
☐	Wed 7 Oct	Connectivity, paths and trees: walks, paths, cycles; connected components, cut vertices, cut edges (bridges), vertex and edge connectivity; Euler and Hamiltonian paths and circuits; trees, forest, spanning trees, Cayley's formula	Gate Smashers Discrete Maths
☐	Thu 8 Oct	Matching, covering and colouring: matching, maximum and perfect matching, Hall's theorem; independent sets, vertex cover, edge cover, cliques and their relations; chromatic number, colouring bipartite/planar/complete graphs, chromatic polynomial; planar graphs, Euler's formula	Gate Smashers Discrete Maths

☐	Fri 9 Oct	Counting: sum and product rules, permutations and combinations with and without repetition, stars and bars, binomial theorem and identities, pigeonhole principle, inclusion-exclusion, derangements, Catalan numbers	Gate Smashers Discrete Maths
☐	Sat 10 Oct	Recurrences and generating functions: forming recurrences from counting problems; solving linear homogeneous and non-homogeneous recurrences (characteristic roots); generating functions: finding coefficients, using them to count and to solve recurrences	Gate Smashers Discrete Maths
☐	Sun 11 Oct	Weekly test: timed PYQs: groups, graph theory, combinatorics, recurrences	GATE Overflow PYQs

Week 3 • 12 - 18 Oct • Programming in C

GA focus (30 min daily): Time, speed, distance and work

✓	Day	Topic and what to cover	Watch / practise
☐	Mon 12 Oct	C fundamentals: data types and sizes, signed/unsigned, type conversion and promotion; operators, precedence and associativity; pre/post increment, short-circuit evaluation, comma operator, bitwise operators and shifts; sizeof	Neso C & Data Structures
☐	Tue 13 Oct	Control flow, functions, scope: if-else, switch fall-through, loops and loop-count questions, break/continue; functions and the call stack; storage classes (auto, static, extern, register); lifetime vs scope; static vs dynamic scoping	Neso C & Data Structures
☐	Wed 14 Oct	Pointers and arrays: pointer arithmetic, pointer-array equivalence, pointer to pointer, arrays of pointers vs pointer to array; 2D arrays, row-major and column-major address calculation; passing arrays to functions	Neso C & Data Structures
☐	Thu 15 Oct	Strings, structures and memory: string functions and output questions; structures, unions, nested structures, self-referential structures; malloc/calloc/free, dangling pointers, memory leaks; call by value vs call by reference; function pointers	Neso C & Data Structures
☐	Fri 16 Oct	Recursion: tracing recursive calls, recursion trees, counting calls and printed output; tail recursion; recursion with static variables; converting recursion to loops; "what does this code compute?" questions: trace an unfamiliar function on small inputs until you see the pattern	Neso C & Data Structures
☐	Sat 17 Oct	Stacks and queues: stack ADT with arrays and linked lists; infix/prefix/postfix conversion and evaluation; balanced parentheses; queues, circular queues, deque, priority queue; queue using two stacks and vice versa	Neso C & Data Structures
☐	Sun 18 Oct	Weekly test: 30 C output questions by hand + stack/queue PYQs	GATE Overflow PYQs

Week 4 • 19 - 25 Oct • Data structures

GA focus (30 min daily): Permutations, combinations, probability (aptitude level)

✓	Day	Topic and what to cover	Watch / practise
☐	Mon 19 Oct	Linked lists: singly, doubly, circular lists; insert, delete, search, reverse; finding middle, detecting loops; merging sorted lists; time complexity of each operation	Neso C & Data Structures
☐	Tue 20 Oct	Binary trees: full, complete, strict, perfect binary trees; max/min nodes for height h, leaf and internal node relations; preorder, inorder, postorder, level order; building a tree from two traversals; expression trees; counting tree shapes	Neso C & Data Structures
☐	Wed 21 Oct	Binary search trees and AVL trees: BST search, insert, delete; resulting tree from an insertion order; number of BSTs with n keys (Catalan); AVL balance factor, LL/RR/LR/RL rotations, min nodes for AVL of height h	Neso C & Data Structures
☐	Thu 22 Oct	Binary heaps: array representation, parent/child indices; min and max heaps; insert, delete-min, heapify, build-heap in O(n); heapsort; k-th smallest; d-ary heaps; valid heap sequences	Neso C & Data Structures
☐	Fri 23 Oct	Graphs as data structures: adjacency matrix vs list and their costs; BFS and DFS traversal order, BFS/DFS trees; time complexity for each representation	Neso C & Data Structures
☐	Sat 24 Oct	Hashing: hash functions, division and multiplication methods; collisions; chaining; open addressing: linear probing, quadratic probing, double hashing; primary and secondary clustering; load factor, expected number of probes	Abdul Bari Algorithms
☐	Sun 25 Oct	Weekly test: timed PYQs: linked lists, trees, BST, heaps, hashing	GATE Overflow PYQs

Week 5 • 26 Oct - 1 Nov • Algorithms I

GA focus (30 min daily): Number series, powers, exponents and logarithms

✓	Day	Topic and what to cover	Watch / practise
☐	Mon 26 Oct	Asymptotic analysis: O , Ω , Θ , o , ω ; comparing growth rates (log, polynomial, exponential, factorial); time complexity of nested and dependent loops; best, worst and average case; space complexity	Abdul Bari Algorithms
☐	Tue 27 Oct	Recurrence relations for algorithms: substitution method, recursion tree method, Master theorem (all cases and when it fails), change of variables	Abdul Bari Algorithms
☐	Wed 28 Oct	Searching and elementary sorts: linear and binary search (and its variants, including bitonic and rotated sorted arrays); bubble, selection, insertion sort; number of comparisons and swaps; stability, in-place, adaptivity; inversions	Gate Smashers DAA
☐	Thu 29 Oct	Merge sort, quicksort, heapsort: recurrences and best/worst cases; partition schemes, pivot choice, quicksort on sorted input; merge procedure comparisons; heapsort analysis	Abdul Bari Algorithms
☐	Fri 30 Oct	Divide and conquer, non-comparison sorts: finding max-min, power computation, Strassen's matrix multiplication; $\Omega(n \log n)$ comparison-sort lower bound; counting, radix and bucket sort	Abdul Bari Algorithms
☐	Sat 31 Oct	Greedy algorithms: greedy choice property; activity selection, fractional knapsack, job sequencing with deadlines, Huffman coding (code lengths, average bits), optimal merge pattern	Abdul Bari Algorithms
☐	Sun 1 Nov	Weekly test: timed PYQs: complexity, recurrences, sorting, greedy	GATE Overflow PYQs

Week 6 • 2 - 8 Nov • Algorithms II

GA focus (30 min daily): Data interpretation: bar, pie, line charts and tables

✓	Day	Topic and what to cover	Watch / practise
☐	Mon 2 Nov	Dynamic programming basics: overlapping subproblems, optimal substructure; memoisation vs tabulation; Fibonacci, coin change, rod cutting, longest increasing subsequence	Abdul Bari Algorithms
☐	Tue 3 Nov	Classic DP problems I: longest common subsequence (table filling, count of LCS), 0/1 knapsack, subset sum, edit distance	Abdul Bari Algorithms
☐	Wed 4 Nov	Classic DP problems II: matrix chain multiplication (number of multiplications, parenthesisation), optimal binary search tree, Floyd-Warshall all-pairs shortest paths	Abdul Bari Algorithms
☐	Thu 5 Nov	Graph traversal applications: DFS discovery/finish times, tree/back/forward/cross edges; cycle detection; topological sort; connected and strongly connected components; articulation points and bridges; bipartite check with BFS	Abdul Bari Algorithms
☐	Fri 6 Nov	Minimum spanning trees: Kruskal (union-find) and Prim; cut and cycle properties; uniqueness of MST, number of MSTs, effect of changing edge weights; complexity with different data structures	Abdul Bari Algorithms
☐	Sat 7 Nov	Shortest paths: Dijkstra (and why negative edges break it), Bellman-Ford and negative cycles, single-source shortest paths in DAGs; complexity with heaps; relaxation order questions	Abdul Bari Algorithms
☐	Sun 8 Nov	Weekly test: timed PYQs: DP, graph algorithms, MST, shortest paths	GATE Overflow PYQs

Week 7 • 9 - 15 Nov • Operating System I

GA focus (30 min daily): Mensuration and geometry

✓	Day	Topic and what to cover	Watch / practise
☐	Mon 9 Nov	Processes and system calls: user vs kernel mode, system calls, interrupts and traps; process states and transitions, PCB, context switch; fork(), exec(), wait() and counting processes/printed lines from fork code	Gate Smashers OS
☐	Tue 10 Nov	Threads and IPC: threads vs processes, user-level vs kernel-level threads, multithreading models; what threads share; inter-process communication: shared memory, message passing, pipes	Gate Smashers OS
☐	Wed 11 Nov	CPU scheduling: FCFS, SJF, SRTF, priority (pre-emptive and not), round robin (effect of quantum), multilevel queue and feedback, HRRN; turnaround, waiting and response time, throughput, CPU utilisation with I/O bursts; starvation, convoy effect	Gate Smashers OS
☐	Thu 12 Nov	Concurrency and the critical-section problem: race conditions; mutual exclusion,	Gate Smashers OS

	Nov	progress and bounded waiting; lock variables, strict alternation, Peterson's solution; hardware support: test-and-set, swap; checking whether given code satisfies the requirements	
☐	Fri 13 Nov	Semaphores and classic problems: binary and counting semaphores, mutex, busy waiting vs blocking; producer-consumer, readers-writers, dining philosophers; monitors; final semaphore values and possible outputs of concurrent code	Gate Smashers OS
☐	Sat 14 Nov	Deadlock: four necessary conditions; resource allocation graphs; prevention; avoidance with safe states and Banker's algorithm; detection and recovery; minimum resources to guarantee no deadlock	Gate Smashers OS
☐	Sun 15 Nov	Weekly test: timed PYQs: fork, scheduling, synchronisation, deadlock	GATE Overflow PYQs

Week 8 • 16 - 22 Nov • Operating System II + Linear Algebra

GA focus (30 min daily): Grammar: tenses, articles, prepositions, subject-verb agreement

✓	Day	Topic and what to cover	Watch / practise
☐	Mon 16 Nov	Memory management and paging: logical vs physical addresses, fixed and variable partitioning, first/best/worst fit, internal and external fragmentation, compaction; paging, page table size, address translation bit splits	Gate Smashers OS
☐	Tue 17 Nov	Advanced paging and segmentation: multi-level page tables, inverted page tables; TLB and effective memory access time; segmentation and segmented paging	Gate Smashers OS
☐	Wed 18 Nov	Virtual memory: demand paging, page faults and effective access time with page-fault service time; FIFO, optimal, LRU, clock replacement; Belady's anomaly; frame allocation, thrashing, working set, locality	Gate Smashers OS
☐	Thu 19 Nov	File systems and disk scheduling: directory structures; contiguous, linked and indexed allocation; inode with direct and indirect blocks and maximum file size; free-space management (bitmap, linked list); disk structure and access time (seek, rotational latency, transfer time, capacity); FCFS, SSTF, SCAN, C-SCAN, LOOK, C-LOOK	Gate Smashers OS
☐	Fri 20 Nov	Linear algebra I: matrix types (symmetric, skew-symmetric, orthogonal, idempotent, nilpotent, triangular); determinants and their properties; rank, row echelon form; systems of linear equations: consistency, unique/infinite/no solution, Gaussian elimination	GO Classes Linear Algebra
☐	Sat 21 Nov	Linear algebra II: eigenvalues and eigenvectors, characteristic equation, properties (trace, determinant, powers, inverse, triangular/symmetric matrices); algebraic vs geometric multiplicity and diagonalisability; Cayley-Hamilton theorem; LU decomposition	GO Classes Linear Algebra
☐	Sun 22 Nov	Weekly test: timed PYQs: memory management, virtual memory, file systems, linear algebra	GATE Overflow PYQs

Week 9 • 23 - 29 Nov • Calculus + Digital Logic I

GA focus (30 min daily): Vocabulary: word usage, synonyms, idioms and phrases

✓	Day	Topic and what to cover	Watch / practise
☐	Mon 23 Nov	Calculus I: limits and standard limits, indeterminate forms and L'Hôpital's rule; continuity and differentiability, checking at a point, piecewise functions	GATE Wallah Engineering Maths
☐	Tue 24 Nov	Calculus II: increasing/decreasing functions, local and global maxima and minima; Rolle's and Lagrange's mean value theorems; definite and indefinite integration, properties of definite integrals, integration by parts, area under a curve	GATE Wallah Engineering Maths
☐	Wed 25 Nov	Number representation and arithmetic: base conversion; sign-magnitude, 1's and 2's complement, ranges and overflow detection; fixed-point representation; BCD and Gray codes; byte ordering (little- vs big-endian); binary addition, subtraction, multiplication (Booth's algorithm)	Neso Digital Electronics
☐	Thu 26 Nov	Floating point and Boolean algebra: IEEE 754 single and double precision, bias, normalised vs denormalised numbers, special values, range and precision; Boolean laws, duality, minterms and maxterms, SOP and POS, functionally complete sets, NAND/NOR realisation	Neso Digital Electronics
☐	Fri 27 Nov	Minimisation: algebraic simplification; K-maps up to 4-5 variables with don't-cares; prime and essential prime implicants; tabular (Quine-McCluskey) method worked fully by hand with prime implicant charts; counting Boolean functions (self-dual, neutral)	Neso Digital Electronics
☐	Sat 28 Nov	Design of combinational circuits: half and full adders/subtractors, ripple-carry and carry-look-ahead adders (delays), comparators, multiplexers and implementing functions with MUX, decoders, encoders and priority encoders, circuit delay questions	Neso Digital Electronics

☐	Sun 29 Nov	Weekly test: timed PYQs: calculus, number representation, minimisation, combinational circuits	Neso GATE Solved Problems, Digital
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Week 10 • 30 Nov – 6 Dec • Digital Logic II + COA I

GA focus (30 min daily): Reading comprehension and inference

✓	Day	Topic and what to cover	Watch / practise
☐	Mon 30 Nov	Sequential circuit elements: latches vs flip-flops; SR, D, JK, T flip-flops: characteristic and excitation tables; flip-flop conversions; race-around condition, master-slave; setup and hold time	Neso Digital Electronics
☐	Tue 1 Dec	Design of sequential circuits: synchronous and asynchronous (ripple) counters, mod-N counters, ring and Johnson counters, counter sequence questions; shift registers; finite state machines (Mealy and Moore), state tables and minimum flip-flops	Neso Digital Electronics
☐	Wed 2 Dec	Instruction set and addressing modes: instruction formats (0-, 1-, 2-, 3-address), expanding opcodes and counting instructions; immediate, direct, indirect, register, register-indirect, indexed, base, relative, auto-increment/decrement modes; RISC vs CISC; instruction cycle	Gate Smashers COA
☐	Thu 3 Dec	ALU, datapath and control unit: ALU design, register transfer and micro-operations, single-bus and multi-bus datapaths, control signals; hardwired control; microprogrammed control, horizontal vs vertical microinstructions, control word and control memory size; practise older GATE PYQs on microprogramming (CSE 1996, 1997, 1999, 2002, 2004, 2013; IT 2004–2008)	Gate Smashers COA
☐	Fri 4 Dec	Memory interfacing and hierarchy: memory chips and organisation, number of chips needed, address decoding; memory interleaving (practise the 2023 PYQ and textbook chip-count problems); hierarchy performance: hit ratio, average access time (hierarchical vs simultaneous access)	Gate Smashers COA
☐	Sat 5 Dec	Cache memory mapping: direct, fully associative and set-associative mapping; tag/index/offset bits, tag memory size; LRU and FIFO replacement, hit/miss sequences; write-through vs write-back, write-allocate; multi-level caches; compulsory, capacity and conflict misses	Gate Smashers COA
☐	Sun 6 Dec	Weekly test: timed PYQs: sequential circuits, addressing modes, control unit, cache	GATE Overflow PYQs

Week 11 • 7 – 13 Dec • COA II + Probability + Databases I

GA focus (30 min daily): Narrative sequencing and critical reasoning

✓	Day	Topic and what to cover	Watch / practise
☐	Mon 7 Dec	I/O interface: programmed I/O, interrupt-driven I/O, vectored interrupts, interrupt priority and daisy chaining; DMA: burst, cycle-stealing and transparent modes; percentage of time the CPU is blocked	Gate Smashers COA
☐	Tue 8 Dec	Instruction pipelining: pipeline stages, speedup, efficiency, throughput; unequal stage delays and buffer delay; structural, data (RAW, WAR, WAW) and control hazards; stalls and CPI; operand forwarding; branch penalty, branch prediction, delayed branch	Gate Smashers COA
☐	Wed 9 Dec	Probability I: sample spaces, axioms, conditional probability, independence, total probability theorem, Bayes theorem; random variables: PMF, PDF, CDF; expectation, linearity of expectation, variance	GATE Wallah Engineering Maths
☐	Thu 10 Dec	Probability II and statistics: uniform, binomial, Poisson, exponential (memoryless) and normal distributions with their mean and variance; standard normal; mean, median, mode, standard deviation; basic correlation	GATE Wallah Engineering Maths
☐	Fri 11 Dec	ER model: entities, attributes (simple, composite, multivalued, derived, key), weak entities; relationships, degree, cardinality, participation; ER-to-relational mapping and the minimum number of tables	Gate Smashers DBMS
☐	Sat 12 Dec	Relational algebra and tuple calculus: keys (super, candidate, primary, foreign); selection, projection, rename, union, difference, Cartesian product, joins (natural, theta, outer), division; number of tuples in results; tuple relational calculus and safe queries	Gate Smashers DBMS
☐	Sun 13 Dec	Weekly test: timed PYQs: DMA, pipelining, probability, ER, relational algebra	GATE Overflow PYQs

Week 12 • 14 – 20 Dec • Databases II

GA focus (30 min daily): Analytical: deduction, induction, syllogisms

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✓	Day	Topic and what to cover	Watch / practise
☐	Mon 14 Dec	SQL: DDL and DML; SELECT with WHERE, GROUP BY, HAVING, ORDER BY; joins; nested and correlated subqueries; IN, EXISTS, ANY, ALL; NULL semantics and aggregates with NULLs; views; predicting query output	Gate Smashers DBMS
☐	Tue 15 Dec	Integrity constraints and functional dependencies: domain, key, entity and referential integrity, ON DELETE CASCADE/SET NULL; FDs, Armstrong's axioms, attribute closure, finding all candidate keys, prime attributes, equivalence of FD sets, minimal (canonical) cover; counting super keys from candidate keys	Gate Smashers DBMS
☐	Wed 16 Dec	Normal forms and decomposition: 1NF, 2NF, 3NF, BCNF and identifying the highest normal form; lossless-join test, dependency preservation; decomposing into 3NF and BCNF	Gate Smashers DBMS
☐	Thu 17 Dec	File organisation and indexing: records, blocking factor, spanned vs unspanned; primary, clustering and secondary indexes; dense vs sparse; multilevel indexes; block accesses needed; B-tree and B+ tree order calculation, insertion and splits, max/min keys and levels	Gate Smashers DBMS
☐	Fri 18 Dec	Transactions: ACID properties, transaction states; schedules; conflict serialisability with precedence graphs, view serialisability; recoverable, cascadeless and strict schedules	Gate Smashers DBMS
☐	Sat 19 Dec	Concurrency control: shared and exclusive locks; two-phase locking (basic, strict, rigorous) and what each guarantees; deadlock in locking, wait-die and wound-wait; timestamp ordering protocol, Thomas write rule	Gate Smashers DBMS
☐	Sun 20 Dec	Weekly test: timed PYQs: SQL, FDs and normal forms, indexing, B+ trees, transactions	GATE Overflow PYQs

Week 13 · 21 - 27 Dec · Computer Networks (2027 syllabus)

GA focus (30 min daily): Analogy, numerical relations, blood relations, coding-decoding

✓	Day	Topic and what to cover	Watch / practise
☐	Mon 21 Dec	Layering, switching and performance: OSI and TCP/IP layers and their jobs; circuit, packet and virtual-circuit switching; transmission, propagation, queuing and processing delay; bandwidth-delay product, RTT, throughput; stop-and-wait efficiency	Gate Smashers CN
☐	Tue 22 Dec	Data link layer: error detection: parity (single and 2D), checksum, CRC (division, detecting burst errors), Hamming distance and detection capability; Hamming code: placing parity bits, correcting a single-bit error	Gate Smashers CN
☐	Wed 23 Dec	Medium access control and Ethernet: pure and slotted ALOHA efficiency; CSMA, CSMA/CD, minimum frame size, binary exponential backoff; probability of success or collision in a slot, expected number of attempts; Ethernet frame format and addressing	Gate Smashers CN
☐	Thu 24 Dec	IPv4: IPv4 header; classful addressing basics, CIDR notation, subnetting and supernetting, number of hosts, longest-prefix match; fragmentation (offset, MF flag, number of fragments); network address translation	Gate Smashers CN
☐	Fri 25 Dec	Routing: distance-vector routing (Bellman-Ford, count-to-infinity, split horizon) and link-state routing (Dijkstra, flooding of link-state packets); computing routing tables	Gate Smashers CN
☐	Sat 26 Dec	TCP, sockets, DNS, HTTP: TCP header, three-way handshake, sequence and acknowledgement numbers; sliding window, go-back-N and selective repeat window sizes; flow control (receiver window); congestion control: slow start, AIMD, fast retransmit, threshold questions; socket API call sequence; DNS resolution; HTTP persistent vs non-persistent. Skip UDP, ARP, DHCP, ICMP, SMTP, FTP: removed for 2027	Gate Smashers CN
☐	Sun 27 Dec	Weekly test: timed CN numericals + PYQs on 2027 topics only	GATE Overflow PYQs

Week 14 · 28 Dec - 3 Jan · Theory of Computation

GA focus (30 min daily): Spatial: translation, rotation, scaling, mirroring

✓	Day	Topic and what to cover	Watch / practise
☐	Mon 28 Dec	Finite automata: designing DFAs (divisibility, substrings, counting), product construction; NFA and ϵ -NFA; subset construction and state counts; Mealy and Moore machines	Gate Smashers TOC
☐	Tue 29 Dec	Minimisation and regular expressions: DFA minimisation (equivalence classes, Myhill-Nerode); regular expressions and identities; regex $\leftrightarrow$ FA conversion, Arden's theorem; regular grammars	Gate Smashers TOC
☐	Wed 30 Dec	Regular languages: closure properties (union, intersection, complement, concatenation, star, reversal, homomorphism, quotient); decision properties; pumping	Gate Smashers TOC

		lemma; spotting regular vs non-regular languages	
☐	Thu 31 Dec	Context-free grammars: derivations and parse trees, ambiguity; simplification (useless, ϵ and unit productions); Chomsky and Greibach normal forms; writing grammars for languages	Gate Smashers TOC
☐	Fri 1 Jan	Push-down automata and CFLs: PDA acceptance by final state and empty stack; deterministic vs non-deterministic PDA, DCFLs; CFL closure properties (not intersection or complement); pumping lemma for CFLs; identifying CFL vs non-CFL	Gate Smashers TOC
☐	Sat 2 Jan	Turing machines and undecidability: Turing machines and variants; recursive vs recursively enumerable languages and their closure; halting problem, reductions, Rice's theorem, Post correspondence problem; decidability table for regular, CFL, recursive and RE problems	Gate Smashers TOC
☐	Sun 3 Jan	Weekly test: timed TOC PYQs; write the closure and decidability tables from memory	GATE Overflow PYQs

Week 15 · 4 - 10 Jan · Compiler Design

GA focus (30 min daily): Paper folding, cutting, assembling, grouping, 2D/3D patterns

✓	Day	Topic and what to cover	Watch / practise
☐	Mon 4 Jan	Compiler phases and lexical analysis: phases and what each produces, symbol table, errors per phase; tokens, lexemes, patterns; counting tokens in code; regular expressions to DFA for lexers	Gate Smashers Compiler Design
☐	Tue 5 Jan	Top-down parsing: eliminating left recursion, left factoring; recursive descent; FIRST and FOLLOW sets; LL(1) table construction and conflicts; checking whether a grammar is LL(1)	Gate Smashers Compiler Design
☐	Wed 6 Jan	Bottom-up parsing: shift-reduce parsing, handles, viable prefixes; operator-precedence relations and parsing a string; LR(0), SLR(1), LALR(1), CLR(1) items and tables; shift-reduce and reduce-reduce conflicts; number of states; power relations between parsers	Gate Smashers Compiler Design
☐	Thu 7 Jan	Syntax-directed translation: synthesised and inherited attributes; S-attributed and L-attributed definitions; annotated parse trees; evaluation order; SDTs that print or compute values	Gate Smashers Compiler Design
☐	Fri 8 Jan	Runtime environments and intermediate code: activation records, stack and heap allocation, static vs dynamic scoping, parameter passing (value, reference, value-result, name); three-address code, quadruples and triples, SSA form, DAG for expressions, backpatching for Boolean expressions and jumps; basic blocks and control-flow graphs	Gate Smashers Compiler Design
☐	Sat 9 Jan	Optimisation and data-flow analysis: local optimisation: constant folding, copy propagation, dead-code elimination, strength reduction, algebraic simplification; data-flow analyses: constant propagation, liveness analysis (live variables, register needs), common subexpression elimination via available expressions	Gate Smashers Compiler Design
☐	Sun 10 Jan	Full-syllabus diagnostic mock: 3-hour full mock + analysis; list your five weakest topics	GATE Overflow PYQs

Phase 2 — Revise (11 - 17 Jan)

Week 16 · 11 - 17 Jan · Revision with subject tests

✓	Day	Task
☐	Mon 11 Jan	Revise General Aptitude + Discrete Maths; subject test on each
☐	Tue 12 Jan	Revise Linear Algebra, Calculus, Probability + Digital Logic; subject tests
☐	Wed 13 Jan	Revise C, Data Structures + Algorithms; subject tests
☐	Thu 14 Jan	Revise Operating System + COA; subject tests
☐	Fri 15 Jan	Revise Databases + Computer Networks; subject tests
☐	Sat 16 Jan	Revise Theory of Computation + Compiler Design; subject tests
☐	Sun 17 Jan	Full mock 1 + analysis

Phase 3 — Mocks and final revision (18 Jan - exam)

Week 17 · 18 - 24 Jan

✓	Day	Task

☐	Mon 18 Jan	Full mock 2
☐	Tue 19 Jan	Analyse mock 2; re-solve every wrong question
☐	Wed 20 Jan	Weakest topic from your mocks: lecture recap + 30 PYQs
☐	Thu 21 Jan	Full mock 3
☐	Fri 22 Jan	Analyse mock 3
☐	Sat 23 Jan	Second-weakest topic: recap + PYQs
☐	Sun 24 Jan	Full mock 4

Week 18 • 25 - 31 Jan

✓	Day	Task
☐	Mon 25 Jan	Analyse mock 4; third-weakest topic
☐	Tue 26 Jan	GATE 2024 CS paper, timed, with the virtual calculator
☐	Wed 27 Jan	Analyse 2024 paper
☐	Thu 28 Jan	GATE 2025 CS paper, timed
☐	Fri 29 Jan	Analyse 2025 paper
☐	Sat 30 Jan	GATE 2026 CS paper, timed
☐	Sun 31 Jan	Analyse 2026 paper; download admit card

Week 19 • 1 - 6 Feb

✓	Day	Task
☐	Mon 1 Feb	Short notes: Maths + GA formulas
☐	Tue 2 Feb	Short notes: OS, COA, Digital Logic
☐	Wed 3 Feb	Short notes: DBMS, CN, TOC, Compiler, C and Algorithms
☐	Thu 4 Feb	Full error-log read-through; light GA practice
☐	Fri 5 Feb	Rest: check exam-centre route, pack admit card + ID, sleep early
☐	Sat 6 Feb	Exam day if CS is on 6 Feb. If later, repeat the mock → analyse → short-notes cycle until your date

Part 4 — PYQ Analysis: GATE CS 2021-2026 vs the 2027 Syllabus

In GATE CS papers from 2021 to 2026, almost every question falls inside the 2027 syllabus. Only 1 of 525 CS questions clearly depended on a topic the revision removed, and at most 5 more might have. What you need beyond the syllabus text is depth on a small set of things GATE keeps asking, listed below, plus extra practice on the four 2027 additions that recent papers rarely test.

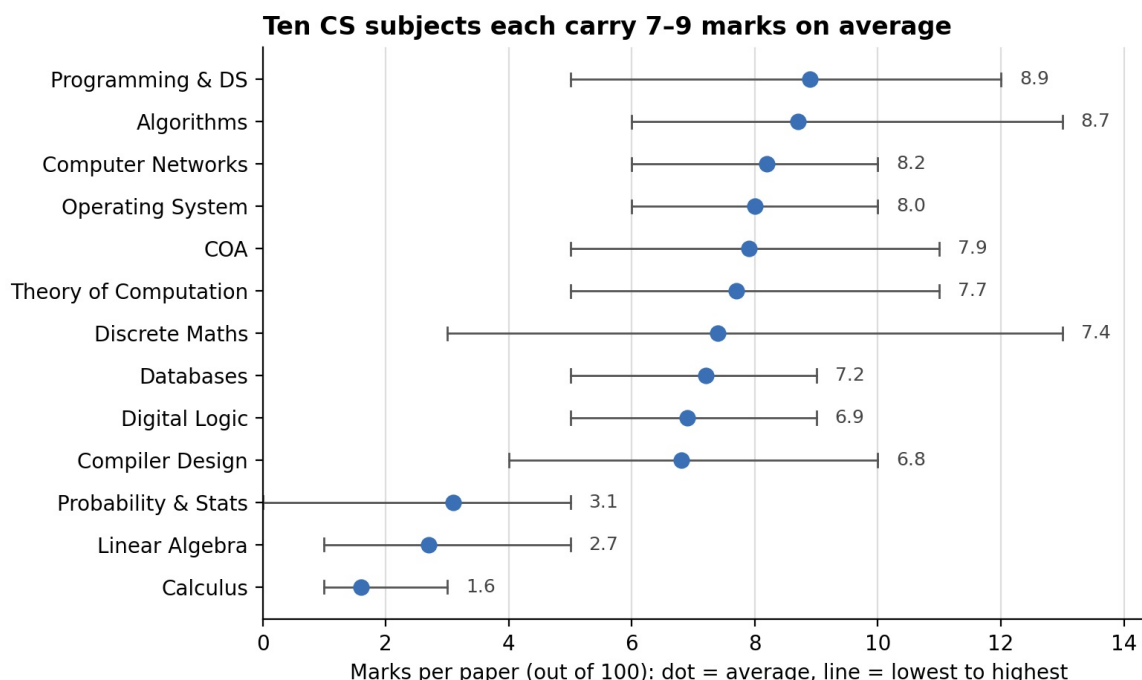
What was analysed

I went through every question in 10 papers, 620 questions in total, using the topic tags on GATE Overflow. Each question was then mapped to a section of the official 2027 syllabus.

Paper	Questions covered
2021 Set 1, 2021 Set 2	65 + 65 (complete)
2022, 2023 (one set each)	65 + 65 (complete)
2024 Set 1	35 of 65 (one page of the question list couldn't be read)
2024 Set 2	65 (complete)
2025 Set 1, 2025 Set 2	65 + 65 (complete)
2026 Set 1, 2026 Set 2	65 + 65 (complete)

Marks per subject use the 9 complete papers only. Topic counts use all 620 questions. Subject totals can differ by 1-2 marks from coaching-institute tables, because some questions sit between two subjects (a heap question can count as Data Structures or Algorithms). As a cross-check, my question-type counts match the published ones for 2023 (34 MCQ, 15 MSQ, 16 NAT) and 2026 Set 2 (11 MSQ, 19 NAT).

Where the marks come from



Average and range of marks per subject

No single CS subject is worth more than about 9 marks on average, so every one of them matters. The engineering maths subjects together average 15 marks, about the same as General Aptitude. Discrete Maths and Algorithms swing the most from paper to paper, by as much as 7-10 marks.

Topics asked almost every year

These 16 topics showed up in at least 7 of the 10 papers. Treat them as near-certain questions: master every question type on

each, not just the concept.

Topic	Subject	Papers (of 10)	Marks (all 10)	What the questions look like
Number representation, 2's complement, IEEE 754	Digital Logic	10	21	Range and overflow, converting to or from IEEE 754 bit patterns, Booth's multiplication
TCP	Computer Networks	10	19	Congestion window after timeouts, sliding-window sizes, sequence numbers, handshake
CPU scheduling	Operating System	10	15	Gantt chart then average waiting/turnaround time; SRTF and round robin most common
C output tracing	Programming & DS	9	26	Pointers, arrays, strings, recursion, static variables: predict the printed value
Pipelining	COA	9	17	Cycles for n instructions with stalls, speedup, hazards and forwarding
Normalisation and FDs	Databases	9	16	Candidate keys, highest normal form, lossless and dependency-preserving decomposition
Cache memory	COA	8	26	Tag/index/offset bits, hit/miss sequences, multi-level average access time
Finite automata	Theory of Computation	8	18	States in the minimal DFA, which strings or languages a DFA accepts
Syntax-directed translation	Compiler Design	7	12	Evaluate an SDT on an input, S- vs L-attributed
Minimum spanning trees	Algorithms	7	15	MST weight, uniqueness, effect of changing edge weights, MST vs shortest path
Time complexity of code	Algorithms	7	12	Complexity of loops or recursive code, comparing growth rates
Binary search trees	Programming & DS	7	12	Tree after insertions, traversal orders, number of possible BSTs
Transactions and concurrency	Databases	7	12	Conflict serialisability, recoverability, 2PL
Boolean algebra and K-maps	Digital Logic	7	10	Minimal SOP, number of prime implicants, functional completeness
Matrices and eigenvalues	Linear Algebra	7	14	Eigenvalue properties, determinant, rank, solutions of linear systems
Context-free languages	Theory of Computation	7	15	Is a language regular, CFL or neither; closure properties

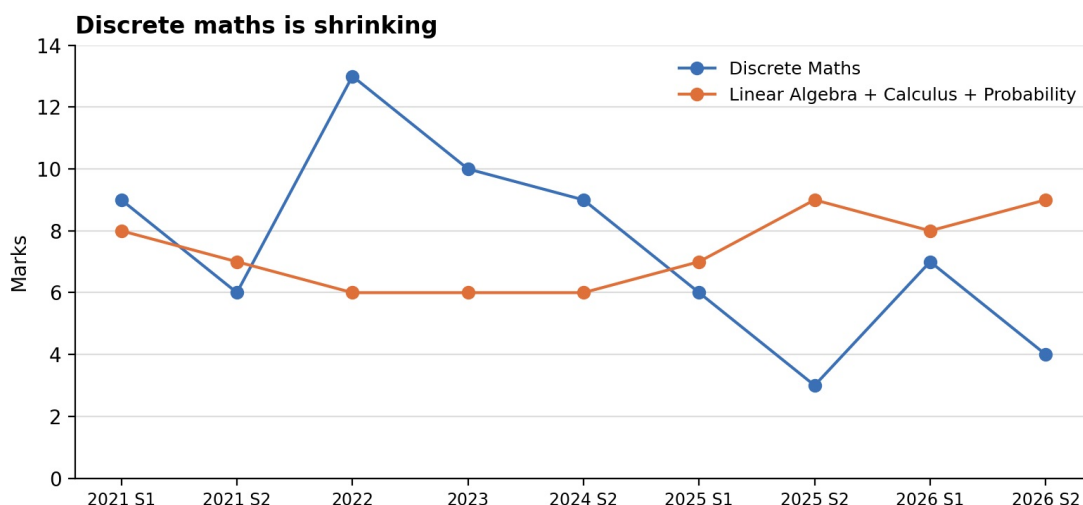
Cache memory and C output tracing carry the most marks of any single topic (26 each across the 10 papers). Both are numerical or tracing questions, so they reward speed built from practice, not reading.

How each subject is asked

Subject	MSQ share	NAT share	What it means for practice
Theory of Computation	44%	13%	Multi-correct statements on closure and decidability. Know the full closure/decidability tables cold; one wrong tick loses the whole question
Operating System	42%	38%	Both statement-checking and numericals. Practise "which of these are true" on synchronisation and paging
Discrete Maths	40%	27%	Statement-heavy on functions, relations, groups and graphs. Learn counter-examples
Digital Logic	37%	21%	Statements about number systems and Boolean functions
COA	17%	48%	Mostly numerical. Drill cache, pipeline and instruction-format calculations with the on-screen calculator
Programming & DS	14%	38%	Tracing code to an exact number. Always dry-run on paper
Probability, Calculus	11-23%	47-54%	Numerical answers; show every step, since there are no options to check against

Shares are of all CS questions in that subject across the 10 papers. There is no negative marking on MSQ or NAT, so never leave those blank.

Trends to plan around



Discrete maths vs other maths, marks per paper

- **Maths is shifting away from discrete maths.** It fell from 13 marks in 2022 to 3–4 in 2025 Set 2 and 2026 Set 2. In all 4 papers since 2025, linear algebra, calculus and probability together outscored discrete maths. Calculus rose to 3 marks in both 2026 papers after 1–2 marks in every earlier paper. Don't skip calculus and probability because they look small.
- **MSQs are rising, unevenly.** 2026 Set 1 had 24 MSQs, the most of any paper analysed, against 11 in 2026 Set 2. Plan for either extreme.
- **COA has grown**, from 5–6 marks in 2021 to 8–11 in 2025–2026, mostly cache and pipelining numericals.
- **Algorithms swings the most** (6 to 13 marks), so a weak algorithms base is the biggest risk on a bad day.

What to learn beyond the syllabus text

The syllabus names broad areas, and GATE asks specific skills inside them that the text never mentions. These 10 came up in recent papers and weren't in the planner yet. Each is now added to the day shown.

Add this	Where it was asked	What to be able to do	Planner day
Byte ordering (little- vs big-endian)	2021 Set 2, 2-mark MSQ	Say which byte sits at which address for a multi-byte value, and read a value back from memory	Wed 25 Nov
Hamming code	2021 Set 1, 2 marks	Place parity bits, find and correct a single-bit error	Tue 22 Dec
Probability applied to networks	2025 Set 1, 2-mark NAT	Success/collision probability for stations sending in a slot, expected number of attempts	Wed 23 Dec
Disk structure and access time	2024 Set 2, 2026 Set 1, both 2-mark NAT	Seek + rotational latency + transfer time, disk capacity, sector addressing. "Secondary storage" left the COA text, but disks are still asked through OS I/O scheduling	Thu 19 Nov
Backpatching	2025 Set 2, 1 mark	Fill jump targets for Boolean expressions and control flow in three-address code	Fri 8 Jan
Operator-precedence parsing	2024 Set 1, 1-mark NAT	Build precedence relations, parse a string step by step	Wed 6 Jan
Eigenvalue multiplicity	2026 Set 1, 1 mark	Algebraic vs geometric multiplicity, when a matrix is diagonalisable	Sat 21 Nov
Counting super keys	2026 Set 1, 2-mark NAT	Count super keys from the candidate keys using inclusion-exclusion	Tue 15 Dec
Modified binary search	2025 Set 2, 2 marks (bitonic array)	Search bitonic and rotated sorted arrays in $O(\log n)$	Wed 28 Oct
"What does this code compute?"	2021 Set 1 and Set 2 (3 questions)	Trace an unfamiliar function on small inputs until you see the pattern (GCD, sum of digits, reversal, and so on)	Fri 16 Oct, plus one per Sunday test

One more pattern to prepare for: **GATE sometimes defines a new term inside the question** and asks you to apply it. 2026 Set 2 defined a Jaccard coefficient on graph neighbourhoods. Read these slowly and try the definition on a tiny example first.

The PYQs also confirm that these less-obvious planner items are worth their day: Booth's multiplication (2025 Set 2), Huffman codes (2021 Set 2), stop-and-wait and sliding-window efficiency (4 papers), double hashing and linear probing (2025, 2026), balls-in-bins counting (2022), symbol tables (2025 Set 1) and graph matching (2026 Set 1).

2027 additions with few recent PYQs

Four topics added or spelled out in the 2027 syllabus were barely tested from 2021 to 2026, so recent papers won't give you enough practice on them. The syllabus now names them explicitly, so prepare them fully anyway, using older papers and textbook exercises.

Topic	Recent PYQs (2021–2026)	Where to practise
Control unit design: hardwired and microprogrammed	None	Older GATE: CSE 1996, 1997, 1999, 2002, 2004 (Q67), 2013 (Q28); GATE IT 2004, 2005 (two questions), 2006, 2008. Control-word size, horizontal vs vertical microinstructions, control memory size
Memory interfacing	1 (2023, 2 marks)	Number of chips for a given memory, address lines, chip-select decoding, interleaving; Hamacher's memory chapter exercises
ALU and datapath design	1 (2025 Set 1, datapath MSQ)	Micro-operations for an instruction on a single-bus datapath, control signals per step
Tabular (Quine–McCluskey) minimisation	None	Work 4–5 functions fully by hand, including prime implicant charts; Morris Mano chapter 3 exercises

Design of sequential circuits, also emphasised in 2027, is already well covered by recent papers: counters or flip-flops appeared in 2021 Set 1, 2023, 2025 Set 1, 2025 Set 2 and 2026 Set 1.

What to skip in old PYQs

The 2027 revision mostly trimmed Computer Networks. When you practise from GATE Overflow, skip questions whose main topic is one of these:

- **Computer Networks:** UDP, ARP, DHCP, ICMP, SMTP, FTP and email, framing, bridging, and flooding as a routing method. In 2021–2026 only one question clearly depended on these (ARP alongside TCP, 2025 Set 2, 1 mark). Five more were tagged only as general network or application-layer protocol questions (2021 Set 1, 2022, 2024 Set 1, both 2026 papers), so check each before skipping it.
- **COA:** questions purely about secondary-storage devices under COA. Keep disk access time and disk scheduling, which are still part of OS.

Everything else in the 2021–2026 papers is still examinable, so all other PYQs from these years are useful practice.

Sources

- GATE Overflow question lists by paper: [2021 Set 1](#), [2021 Set 2](#), [2022](#), [2023](#), [2024 Set 1](#), [2024 Set 2](#), [2025 Set 1](#), [2025 Set 2](#), [2026 Set 1](#), [2026 Set 2](#)
- [GATE Overflow microprogramming questions](#)
- [GATE 2027 CS syllabus](#), IIT Madras
- Cross-checks: [GO Classes 2023–2025 weightage analysis](#), [GeeksforGeeks GATE CSE 2026 Shift 2 analysis](#)